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Frontal-Temporal Synchronization of EEG Signals Quantified by Order Patterns Cross Recurrence Analysis During Propofol Anesthesia

Reza Shalbah, Hamid Behnam, Jamie W. Sleight, D. Alistair Steyn-Ross and Moira L. Steyn-Ross

Abstract— Characterizing brain dynamics during anesthesia is a main current challenge in anesthesia study. Several single channel Electroencephalogram (EEG) -based commercial monitors like the Bispectral index (BIS) have suggested to examine EEG signal. But, the BIS index has obtained numerous critiques. In this study, we evaluate the concentration-dependent effect of the propofol on long-range frontal-temporal synchronization of EEG signals collected from eight subjects during a controlled induction and recovery design. We used order patterns cross recurrence plot and provide an index named order pattern laminarity (OPL) to assess changes in neuronal synchronization as the mechanism forming the foundation of conscious perception. The prediction probability of 0.9 and 0.84 for OPL and BIS specified that the OPL index correlated more strongly with effect-site propofol concentration. Also, our new index makes faster reaction to transients in EEG recordings based on pharmacokinetic and pharmacodynamic model parameters and demonstrates less variability at the point of loss of consciousness (standard deviation of 0.04 for OPL compared with 0.09 for BIS index). The result show that the OPL index can estimate anesthetic state of patient more efficiently than the BIS index in lightly sedated state with more tolerant of artifacts.

Index Terms—Brain states, Cross recurrence plot, Electroencephalogram, General anesthesia, Synchronization.

I. INTRODUCTION

Current studies with multi-channel Electroencephalogram (EEG) [1], transcranial magnetic stimulation [2] and neuroimaging [3-4] have helped notably to our finding of how anesthetics could suppress consciousness. But, these methods are impossible for the usual intra-operative evaluation of depth of anesthesia. Conversely, single channel EEG-based

commercial monitor of anesthesia depth named the bispectral index (BIS; Aspect Medical Systems, Newton, MA) utilize experimentally-derived techniques which are not based in the cause of consciousness [5]. Also, the BIS index has received various disapproval, such as sensitivity to artifacts [6], failure to regain its baseline value after emergence [7] and long signal processing time delays [8-9]. Therefore, recognizing cognitive neuroscience of consciousness which can be measured usually in operating rooms would be a considerable advance [10].

To understand how the anesthetics adjust activity of the brain, it is essential to study not only the brain regions concerned during general anesthesia but also how the connectivity of diverse functional areas of the brain are changed by general anesthesia [2-3]. Functional areas in the brain are interconnected, generating networks which permit complicated processing of internal and/or external stimulus representations. Anesthetics disturb interaction between different cortical regions, and so weaken large-scale integration of information imagined to be a requirement to suitable brain function [4]. Therefore, suppression of communication among cortical areas is imagined to be a key player in the process of unconsciousness.

Synchronization in neuronal networks has identified as an important feature for quantifying the communication between different cortical regions and expresses the functional association of the cortical spatially signals [11]. There are major problems with traditional methods of estimation of synchronization such as cross-correlation technique [12]. This technique might run into crisis when the oscillations are weakly related. Also, synchronization can be quantified from the phase difference of the EEG signals (coherence measure) [13]. But, this method is not adequate to deal with nonlinear dependencies in EEG signals and also it is sensitive to artifact.

Recently, nonlinear synchronization measures such as cross-approximate entropy [14] or harmonic wavelet bicoherence [15] methods have been proposed. However, since the EEG data is non-stationary and noisy, these techniques should be performed with caution. These problems are relatively solved by the cross recurrence plot (CRP) analysis [16]. This analysis has few assumptions and looks to see if two channels are in a similar part of phase space. However, this method is based on calculating distances of

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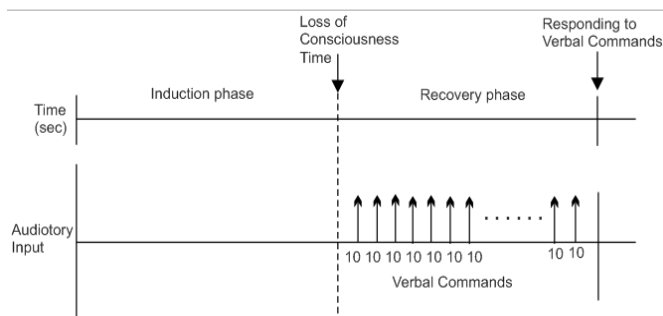


Fig. 1. The diagram of the experimental design. Propofol concentration fixed at 150ml/h.

trajectories, which is extremely sensitive to amplitude ranges and baseline drifts. To overcome these drawbacks an alternative version of CRP based on order patterns was introduced [17] and applied effectively to event-related potentials [18]. Order patterns are used to encode the original EEG signals as a series of symbols of which a cross recurrence plot is calculated. The main advantage of the symbolic representation is the robustness against non-stationary and noisy EEG signals [19]. Such a CRP represent those times, when specific local order patterns in the one signal recur on the same local order patterns of other signal. The information contained in this plot is quantified by measures of complexity based on the vertical structures and is named order pattern laminarity (OPL). This measure is theoretically simple, computationally proficient and captures information about the time dependent behavior of the dynamical system such as brain.

In the current study, we will examine the concentration-dependent effect of the propofol on long-range frontal-temporal synchronization of EEG quantified as OPL index in the human during propofol anesthetic induction and recovery with a regular drug routine. In addition, we evaluate the assumption that this synchronization associate with alteration in a patient's physiologic state in lightly sedated state. For comparison purposes, we will assess a phase synchronization measure and cross-correlation technique which has been recently applied to the study of EEG signals. Also, prediction probability and pharmacokinetic and pharmacodynamic (PKPD) modeling are used to evaluate the effectiveness of OPL index in reflecting effect-site concentration of the anesthetic drug. We compare the new index with the BIS index which extracted from the pre-frontal EEG channel.

II. MATERIALS AND METHODS

A. Subjects and Data Acquisition

8 human volunteers (five male and three female, weight 59-120 kg, age 18-42 yr, American Society of Anesthesiologists physical status I or II) were undergone a short propofol anesthetic. The details have been previously described [20]. All participants gave written informed consent and the protocols used in this study were approved by the hospital

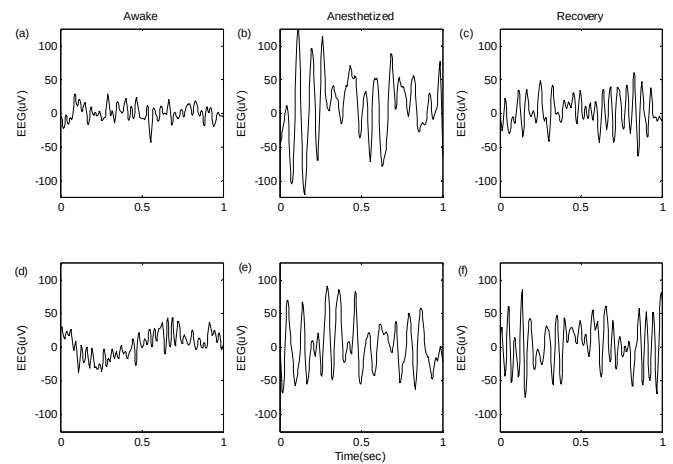


Fig. 2. Examples of 1-s EEG epochs (a-c) at Fp1-F7, (d-f) at C3-T3 when the volunteer is awake (10-s after beginning of study), anesthetized (LOC time) and recovery (response to verbal commands).

ethical committee. A short propofol anesthetic was given to every volunteer in agreement with the Australian and New Zealand College of Anesthesia Guidelines.

Two bipolar EEG signals (Fp1-F7 and C3-T3) were recorded using scalp electrodes. The reference electrode was positioned at Fpz. Alters in EEG activity of the left temporal and prefrontal were discovered using these montages. The raw EEG signals and BIS index were recorded using the A-1000 BIS monitor onto an individual computer. The sampling rate of EEG data and BIS index were 256 Hz and 0.2 Hz respectively. The key notch filter was set at 50 Hz. The high and low frequency filters were arranged at 70 and 0.5 Hz respectively.

The empirical sequence was demonstrated in figure 1. All volunteers obtained continuous infusion of Propofol using a syringe pump after intravenous access. Propofol concentration fixed at 150ml/h (1500mg/h). The BIS index and the two channel EEG signals were recorded at the beginning of the infusion. When the subjects lost consciousness, the induction was stopped. Loss of consciousness (LOC) time was approximated by the time of dropping a water-filled syringe, held between thumb and forefinger. After this time, the volunteers were then permitted to awaken and a pre-recorded tape of verbal commands was started. These commands such as "move your left foot" were at ten-s intervals. The recovery time was approximated when patient was able to give a correct response to verbal commands. 60 s after recovery time, the period research was finished. Figure 2 demonstrates some examples of 1 s EEG epochs when the volunteer was awake (10 s after beginning of study), anesthetized (LOC time) and recovery (response to verbal commands).

B. Phase space reconstruction

The reconstruction of the phase space trajectory of the signal was the first step in the analysis of a signal by means of non-linear dynamics theory. The phase space vectors $\vec{x}_i$ for a one-dimensional time series u_i could be reconstructed via the

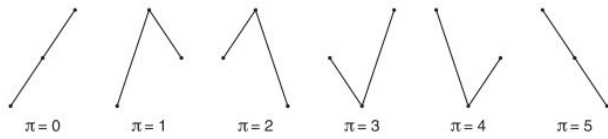


Fig. 3. **Order patterns at embedding dimension $d=3$.** They describe a specific rank order of three data points.

Taken's time delay method $\vec{x}_i = (u_i, u_{i+\tau}, \dots, u_{i+(d-1)\tau})$ with delay τ and embedding dimension d [21]. However, the choice of time delay and embedding dimension were often not well defined.

C. Recurrence plots

The recurrence of states was an essential property of complex dynamical systems [22]. The method of recurrence plot (RP) was presented to visualize time dependent behavior of a dynamical system which could be considered as a trajectory $\vec{x}_i \in \mathbb{R}^d$ ($i = 1, \dots, N$) in the d -dimensional phase space. **The key step of this visualization was the computation of a $N \times N$ matrix**

$$R_{ij}(\varepsilon) = \begin{cases} 1 : \|\vec{x}_i - \vec{x}_j\| \leq \varepsilon \\ 0 : \text{otherwise} \end{cases} \quad \vec{x}_i \in \mathbb{R}^d, i, j = 1, \dots, N \quad (1)$$

where $\|\cdot\|$ was a norm (e.g. Euclidean), (ε) was the cut-off distance and $N = L - (d - 1) * \tau$ (L is length of one dimensional time series).

D. Cross recurrence plot

The cross recurrence plot (CRP) was a bivariate extension of the RP [16]. Assume we had two dynamical systems, each one represented by the trajectories $\vec{x}_i$ and $\vec{y}_j$ in a d -dimensional phase space. Analogous to the RP, the corresponding cross recurrence matrix was defined by:

$$CR_{ij}(\varepsilon) = \begin{cases} 1 : \|\vec{x}_i - \vec{y}_j\| \leq \varepsilon \\ 0 : \text{otherwise} \end{cases} \quad \vec{x}_i \in \mathbb{R}^d, i = 1, \dots, N; \vec{y}_j \in \mathbb{R}^d, j = 1, \dots, N \quad (2)$$

A CRP looked for those times when a state of the first system was within the cut-off distance of the state of the other system [19].

E. Order patterns cross recurrence plot and measures

In this paper, we ignored the actual spatial distance in phase space and, instead, described a recurrence via the local order patterns of a trajectory [17]. We might begin by comparing $d = 2$ time instances using a one-dimensional time series. In this case there were 2 feasible relations between u_i and $u_{i+\tau}$, ignoring equality. We indicated these relations as local order patterns π as

$$\pi_i = \begin{cases} 0 : u_i < u_{i+\tau} \\ 1 : u_i > u_{i+\tau} \end{cases} \quad (3)$$

with the scaling parameter τ . In dimension $d = 3$ there were

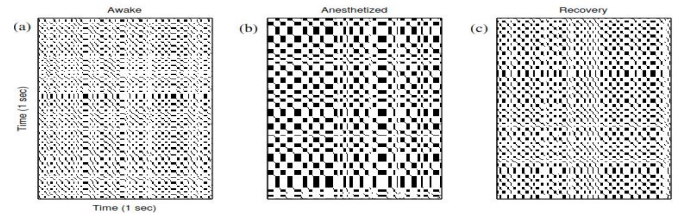


Fig. 4. **Order patterns cross recurrence plot ($d = 4, \tau = 1$) of 1-s EEG epochs of Fig. 2 when the volunteer is (a) Awake (b) Anesthetized and (c) Recovery.** The number of long vertical lines in the anesthetized EEG segment is more than the ones of recovery and awake EEG segments.

$d! = 6$ dissimilar order patterns in the triple $(u_i, u_{i+\tau}, u_{i+2\tau})$ feasible (figure 3). In general the d components in $\vec{x}_i = (u_i, u_{i+\tau}, \dots, u_{i+(d-1)\tau})$ could form $d!$ dissimilar patterns. Using these local order patterns, we formed a novel symbolic time series π_i from each point of phase space trajectories $\vec{x}_i$ as follows:

$$\vec{x}_i = (u_i, u_{i+\tau}, \dots, u_{i+(d-1)\tau}) \rightarrow \pi_i \quad (4)$$

The same theory was applied to produce another symbolic time series φ_j from each point of phase space trajectories $\vec{y}_j$ as follows:

$$\vec{y}_j = (v_j, v_{j+\tau}, \dots, v_{j+(d-1)\tau}) \rightarrow \varphi_j \quad (5)$$

The order patterns cross recurrence plot (OPCRP) was then described by the pair-wise test of local order patterns between two different systems:

$$CR_{ij}(d) = \begin{cases} 1 : \pi_i = \varphi_j \\ 0 : \text{otherwise} \end{cases} \quad i = 1, \dots, N; j = 1, \dots, N \quad (6)$$

Therefore, phase space vectors $\vec{x}_i$ and $\vec{y}_j$ were described as neighboring if their local order patterns coincided: $\pi_i = \varphi_j$.

The information enclosed in an OPCRP could be quantified with measures of complexity based on the recurrence point density and vertical structures [19]. The laminarity measure was based on the vertical lines. The frequency distribution of the lengths v of the vertical structures in the OPCRP was $P^d(v) = \{v_i; i = 1, 2, \dots, N\}$. Therefore, the ratio of recurrence points creating vertical structures to all recurrence points was named the order patterns laminarity (OPL). Its formulation was given below:

$$OPL = 1 - \frac{\sum_{v=v_{\min}}^N v P^d(v)}{\sum_{v=1}^N v P^d(v)} \quad (7)$$

where $v_{\min}$ was the threshold. According to theoretical considerations and other studies, the parameters set $d = 4, \tau = 1$ and $v_{\min} = 4$ were used in this study [17-19]. The synchronization between two EEG channels is quantified with OPL index. The OPL index is compared with other traditional synchronization measures such as cross-correlation technique (COR index) and coherence method (COH index). All indices are computed over a window of 10-s with an overlap of 5 s.

F. Pharmacokinetic-pharmacodynamic modeling

Pharmacokinetic-pharmacodynamic (PKPD) modeling

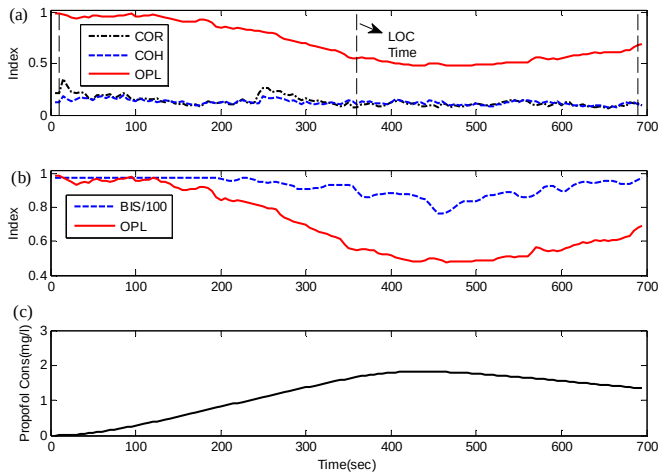


Fig. 5. An example of the EEG measures (a-b) and simultaneous effect-site propofol concentration (c) versus time. From left to right, three dashed vertical lines are used to show the time of awake, loss of consciousness and recovery, respectively. OPL tracks the effect-site drug concentration more quickly (after 140th s) than BIS and is more tolerant of artifacts (about 350th and 430th s). Both COR and COH indices have no significant variation.

described how the concentration of the drug changes the EEG measure [23]. The continuous effect-site concentration (C_{eff}) in the brain for each volunteer was estimated using IVA-SIM computer program [24]. Relationship between EEG measures and C_{eff} was modeled with the generally-used nonlinear sigmoid model:

$$Effect = E_{max} - (E_{max} - E_{min}) * \frac{C_{eff}^{\gamma}}{EC_{50}^{\gamma} + C_{eff}^{\gamma}} \quad (8)$$

where γ interprets the slope of the response-concentration relationship and the E_{max} and E_{min} are the maximum and minimum effect for every volunteer. EC_{50} was the concentration where the effect is midway between the maximum and minimum values. The coefficient of determination R^2 was calculated by

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (9)$$

where $\hat{y}_i$ and y_i were the corresponding model prediction and experimental effect for a definite time, respectively. The $\bar{y}$ was the mean of all experimental effect over time.

G. Statistical Analysis

Correlation between C_{eff} and measured EEG index was assessed by using the prediction probability (P_k) statistics [25], as is frequently utilized in latest publications [26]. The P_k values were determined with the 2003 Excel software [25]. Also, linear regression and Pearson correlation coefficient (corr) were employed as a measure of relationship between OPL and BIS index. In addition, the performance of OPL and BIS index were compared pair-wise by Student t-test.

III. RESULTS

The synchronization between two EEG channels is quantified with OPL index. OPCR of 1-s EEG epochs of figure 2 are shown in figure 4. Visual inspections of OPCR show that the number of long vertical lines in the OPCR of

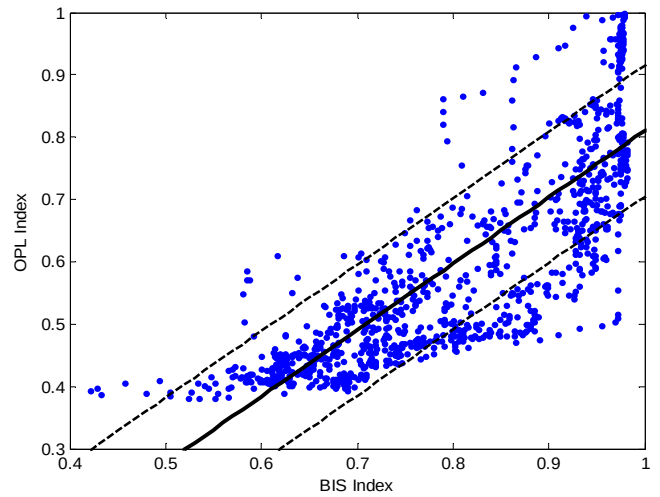


Fig. 6. A scatter plot for the OPL and BIS for all subjects. The best-fit line is bold and dashed lines correspond to the 95% confidence boundaries. This fitted linear relation indicates that the two methods are extremely correlated.

anesthetic EEG segment (LOC time) is more than the ones of recovery and awake EEG segments. Long vertical line means that signals are doing the same thing for some time steps. In this situation, some segments of signals stay in the same phase space region for some time. This occurs whenever the signals are dominated by slow (theta or alpha) waves and high-frequency components of EEG signals are disappeared. Slow waves take longer times, and hence the up-slope patterns and down-slope patterns line up with each other. Therefore during anesthesia, the EEG segments become more synchronized with each other over the time-scales used to construct the cross recurrence maps. Figure 4(b) also shows that we are primarily detecting the propofol-induced increase in alpha/spindles (which are widely synchronous) - as there are about 10 (-15) large black dots /s.

An example of the EEG measures and simultaneous effect-site propofol concentration versus time is demonstrated in figure 5. It is obvious that OPL index follows the gross changes in EEG signals with increasing and decreasing propofol concentration (figure 5(a)). In contrast, both COR and COH indices fail to reliably follow the change in subject states. The range of the recorded BIS index is between 0-100, but is transformed to 0-1 (divided to 100) to make the comparison with OPL index possible (figure 5(b)). According to this plot, OPL tracks the effect-site drug concentration more quickly (after 140th s) representing that OPL has a shorter delay than BIS particularly about the change from consciousness to unconsciousness. Also, the BIS index demonstrates some irregular upward trends about 350th and 430th s, while this deviation cannot be seen in OPL. The limitations of BIS index may be elucidated as its slow preprocessing causes the risk of time delay for a rapid transition from awake to the anesthetised state and often makes substantial loss in the amount of background EEG activity leading the BIS to increase incorrectly. Thus, the OPL index is robust to artifacts of this nature.

A scatter plot for the OPL and BIS is demonstrated in figure

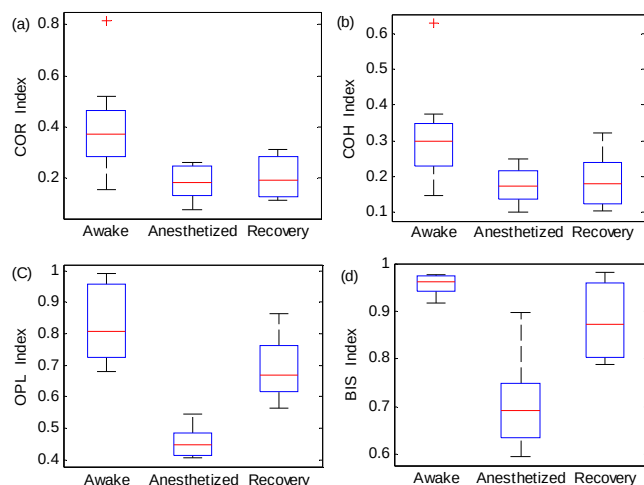


Fig. 7. Box plots of EEG derived measures over 10-s epochs for all subjects. The variability in OPL index is smaller than BIS index at the anesthetized state. Both COR and COH indices cannot differentiate between anesthetized and recovery states.

6 from the volunteers (1081 data points). Linear equation is fitted to all data points with the relation as $OPL = 1.06 \cdot BIS - 0.25$. Dash line shows 95% confidence boundaries around the linear black bold line. Few data points go beyond the 95% confidence boundaries. The high Pearson correlation coefficient ($\text{corr} = 0.82$) demonstrates the similarity of these two techniques for following commonly alters in propofol drug effect.

OPL and BIS values over 10-s epochs at the awake (10 s after beginning of study), anesthetized (LOC time) and recovery (response to verbal commands) states are evaluated for each subject and a box plot is created (figure 7). The OPL values are 0.88 ± 0.12 , 0.46 ± 0.04 , and 0.69 ± 0.1 (mean $\pm$ SD) in the three states, respectively; the corresponding BIS values are 0.95 ± 0.02 , 0.7 ± 0.09 , and 0.88 ± 0.08 . The values of both indices reduce from the awake to the anesthetized state, and subsequently increase at the recovery state. However, the variability in OPL index is considerably smaller than BIS index at the point of LOC. It demonstrates that BIS index does not reliably identify LOC. Rather, OPL index directly reflects the neurophysiologic processes governing conscious state. It should be noted that COR and COH indices cannot differentiate appropriately between anesthetized and recovery states. The reason is that EEG signals at frontal and temporal lobe have different amplitude and noise which make the result of these synchronization measures incorrect.

Using the PKPD modeling, the C_{Eff} can be computed and subsequently, the relation between OPL and BIS values and the C_{Eff} can be achieved for the same subject (figure 8). The results show that the propofol significantly decreases both indices in a concentration-dependent manner. Path-dependent parameters including γ and EC_{50} are then obtained with fitting a non-linear sigmoid E_{max} curve individually to the C_{Eff} versus EEG measures. The modeled parameters are recorded in table 1 as mean $\pm$ SD. The values of γ obtained from OPL

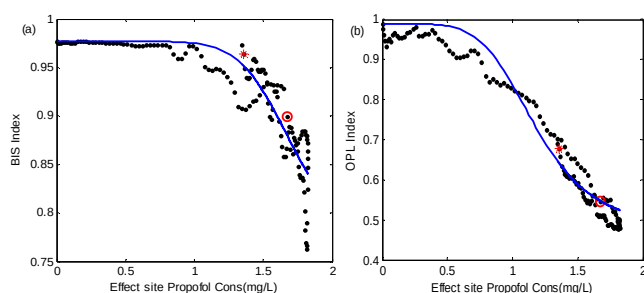


Fig. 8. The relationship between the effect site propofol concentration C_{Eff} and (a) BIS (b) OPL values for the same patient. The circle and the asterisk symbols are used to show the time of LOC and recovery respectively. The goodness-of-fit (R^2) between the OPL and the approximated C_{Eff} is higher than that between BIS and C_{Eff} .

are significantly bigger and EC_{50} smaller than the ones obtained from BIS mostly because of the faster response of OPL to alters in EEG series. Also, the goodness-of-fit (R^2) between the OPL and the approximated C_{Eff} (0.85 ± 0.1) is significantly higher than that between BIS and C_{Eff} (0.80 ± 0.13). The difference is statistically significant ($p < 0.001$).

To reveal whether EEG measures can properly estimate C_{Eff} , the prediction probability P_k is computed for each subject. The P_k values of COR, COH, BIS and OPL are correspondingly 0.7 ± 0.28 , 0.68 ± 0.19 , 0.84 ± 0.13 and 0.9 ± 0.07 . This difference between OPL with other indices are statistically significant ($p < 0.001$, paired t-test) and implies that OPL has a convincing ability to follow the propofol effect-site concentrations over the whole range of data points.

IV. DISCUSSION AND CONCLUSION

Characterizing brain dynamics with multisite EEG recordings is important for understanding the neurophysiology of general anesthesia and for developing more principled strategies for monitoring the brain states during medical procedures [1, 27]. We find a propofol-induced increase in non-linear synchronization between pairs of EEG channels at frontal and temporal regions, implying that the EEG waves including all frequency bands become more synchronized under anesthesia. It should be noted that when the OPL index is decreased it means that synchronization is increased. This study reveals that OPL index can estimate anesthetic state of patient more efficiently than the BIS index in lightly sedated state.

We find a propofol-induced increase in synchronization between separated channel pairs under anesthesia. Anesthetics may force independent cortical subnetworks into more synchronous, uniform activity patterns [28]. Furthermore, input from long-range connections from other cortical areas is weakened under anesthesia [2-3]. It looks that development of GABAergic currents, which bias synaptic interaction towards inhibition and the deterioration of long-range excitatory synaptic inputs by propofol anesthetics cause to more synchronization between frontal and temporal regions [29]. In addition, In the case of light propofol anaesthesia, the

predominant slower waveform is the alpha frequency "sleep spindle"-like pattern [30]. Since this is generated - or at least coordinated - in the thalamus, and therefore is absolutely synchronous across widely spaced cortical areas, it is most likely to be the main influence on the observed increase in synchrony.

The results show that anesthetics encourage alterations of thalamocortical networks and slow principal frequencies. Anesthetics impair brain function by making equality of signals and impeding an exchange of independent information between cortical areas [2-4]. A number of experimental investigations are in agreement with this concept. Volatile anesthetics disturbed anterior-posterior phase synchronization of field responses of rats stimulated by light flashes [31]. Another animal study demonstrates that volatile anesthetics reduce the independence of spontaneous EEG signals recorded from diverse hemispheres of rats [32]. In humans, a variety of general anesthetics decouple gamma rhythms between homologous areas in diverse hemispheres as well as between anterior and posterior cortical areas during the transition from waking to loss of consciousness [33]. Also, a considerable reduction in feedforward and feedback connectivity in the anesthetized state using symbolic transfer entropy method has discovered [34]. In addition, a breakdown of cortical effective connectivity is revealed using transcranial magnetic stimulation during midazolam anesthesia [2].

The statistical analysis exhibits several benefits of OPL over BIS index. First, the computation of BIS value is very complicated and needs more computation time than our method [7-9, 35]. Rather, our proposed measure needs just basic computation like sorting and comparative which can be calculated quickly. Second, the rejection of artifacts in BIS is extremely difficult and causes the risk of time delay for a rapid transition from awake to the anesthetised state or vice versa [8, 36]. Without complicated artifact removal step, our proposed method is more tolerant of artifacts, as revealed from clinical observations (figure 5) and theoretically. In the OPL index, we translate a given EEG series into symbolic sequences. These sequences are robust regarding the all types of artifacts in EEG data. Third, the variability in OPL index is noticeably smaller than BIS index at the point of LOC (figure 7). It reveals the fact that OPL index as depth of anaesthesia measure consistently recognizes LOC. Rather, BIS index is surrogate measures that correlate roughly with the actual neurophysiological processes governing conscious state. Forth, OPL measure causes faster reaction to transients in volunteers' consciousness levels as demonstrated in the PKPD modeled parameters (table 1) and also the clinical observation (figure 5). Finally, through the prediction probability statistics P_k , it is revealed that the OPL index correlated more closely with the C_{Eff} than the BIS index.

The following two points should be mentioned as areas to be further explored. First, although the assessment existing in the paper with data from 8 subjects anesthetized with propofol looks limited, the statistical significance found with these volunteers verifies that the differences must be relatively robust. Also we admit that this research is mainly a proof-of-

TABLE I
PKPD PARAMETERS FOR OPL AND BIS

Parameter	OPL	BIS
γ	5.82 ± 1.65	4.67 ± 1.98
EC_{50}	0.83 ± 0.22	1.16 ± 0.21
R^2	0.85 ± 0.10	0.80 ± 0.13
$E_{\max}$	0.86 ± 0.09	0.96 ± 0.02
$E_{\min}$	0.42 ± 0.05	0.66 ± 0.08

principle and the real clinical significance will just be concluded with future larger trials of the some techniques of analysis. Second, the puzzling effects of surgical stimulation or deep level of anesthesia are not incorporated in the collected data. This drawback should be solved in future studies by representing a database which is gathered at surgical levels of anesthesia and deep level of anesthesia including burst suppression pattern or delta wave.

In conclusion, synchronization between pairs of EEG channels at frontal and temporal lobe quantified as OPL index is increased in 8 subjects receiving intravenous anesthetics and is decreased upon recovery in a usual clinical setting. The goodness-of-fit of the pharmacokinetic- pharmacodynamic model, prediction probability and variability at the point of loss of consciousness using the OPL index ($R^2 = 0.85$, $P_k = 0.9$, $SD = 0.04$) was better than that using the BIS index ($R^2 = 0.80$, $P_k = 0.84$, $SD = 0.09$). This investigation characterizes a key translational step from the cognitive neuroscience of consciousness to more complicated monitoring of anesthetic effects in volunteers.

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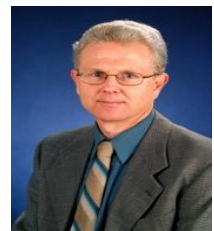
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